

University of Parma
Department of Engineering and Architecture
Degree Program in Computer Engineering
Curriculum: Artificial Intelligence

Deep Learning and Generative Models

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Image_Captioning_CNN_LSTM

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Abstract

This report presents the implementation and evaluation of a CNN-LSTM image captioning system trained on Flickr8k. The project covers the complete pipeline required by the assignment: caption preprocessing, vocabulary construction, training/validation file preparation, supervised training, BLEU-based evaluation, and qualitative analysis. A controlled experimental comparison was conducted on four configurations: baseline, removal of scheduled sampling, removal of label smoothing, and CNN fine-tuning. The best result was obtained by the fine-tuning configuration, which achieved BLEU-4 equal to 0.0649, outperforming the baseline BLEU-4 score of 0.0472.

1 Introduction

Image captioning is a multimodal problem in which a system receives an image and must produce a natural language description of its content. The task combines visual recognition and language generation, and can therefore be naturally formulated as an image-to-sequence problem.

The objective of this project was to implement a complete CNN-LSTM captioning pipeline in accordance with the course assignment. Beyond the model itself, the work also focused on making the entire workflow reproducible, from data preparation to systematic experimental comparison.

2 Assignment Requirements

The assignment required:

- the use of Flickr8k;
- a CNN-LSTM architecture;
- caption cleaning;
- vocabulary construction and export;
- pairing image features with tokenized captions during training;
- word prediction through the decoder output during testing.

The final repository satisfies these requirements through a modular and reproducible implementation.

3 Dataset and Preprocessing

3.1 Flickr8k

Flickr8k associates each image with five reference captions. In the final project, the dataset was transformed into reproducible artifacts:

- `data/vocab.json`

- `data/Flickr8k_text/train.csv`
- `data/Flickr8k_text/val.csv`
- `data/Flickr8k_text/test_captions.json`

Final observed sizes:

- 30000 training caption rows
- 5000 validation caption rows
- 1000 test images
- 5 reference captions per image

3.2 Caption Cleaning

Captions were preprocessed using:

- lowercase conversion
- punctuation removal
- digit removal
- single-letter word removal
- whitespace normalization

The final vocabulary contains 2985 tokens, including the special tokens `<pad>`, `<bos>`, `<eos>`, and `<unk>`.

4 Model Architecture

The system follows an encoder-decoder design.

4.1 Encoder

The encoder uses pretrained InceptionV3. The classification head is removed and the extracted features are projected into the decoder embedding space.

4.2 Decoder

The decoder is a single-layer LSTM. During training, it receives the image-conditioned representation together with the tokenized caption. A final linear layer maps hidden states to vocabulary logits for next-token prediction.

4.3 Training Features

The implemented training loop supports:

- scheduled sampling
- label smoothing
- gradient clipping
- checkpoint saving
- checkpoint resume
- BLEU-1, BLEU-2, BLEU-3, BLEU-4 evaluation

5 Implementation Details

The final repository is organized as follows:

- `src/vocab_builder.py`: cleaning, vocabulary construction, split export
- `src/prepare_data.py`: convenience data preparation script
- `src/dataset.py`: dataset loading and batching
- `src/model.py`: CNN-LSTM definition
- `src/train.py`: training loop and checkpoint logic
- `src/eval.py`: caption generation and BLEU evaluation
- `src/visualize.py`: qualitative visualization
- `src/ui_streamlit.py`: optional interactive interface

During revision, several structural issues were corrected, including:

- incorrect parsing of Flickr8k captions
- missing train/validation/test artifacts
- incorrect next-token loss alignment
- ineffective encoder fine-tuning

6 Experimental Setup

All final experiments were executed in the `captioning` environment with PyTorch 2.9.1+cpu. The device used for training and evaluation was CPU.

Compared configurations:

1. Baseline
2. No Scheduled Sampling
3. No Label Smoothing
4. Fine-tuning

All final runs were executed for 5 epochs.

7 Quantitative Results

Table 1: BLEU comparison across the final experiments.

Experiment	BLEU-1	BLEU-2	BLEU-3	BLEU-4
Baseline	0.2553	0.1556	0.0928	0.0472
No Scheduled Sampling	0.2232	0.1242	0.0717	0.0396
No Label Smoothing	0.2404	0.1435	0.0809	0.0394
Fine-tuning	0.3301	0.2022	0.1189	0.0649

Table 2: Training summary.

Experiment	Best Val Loss	Final Train Loss	Final Val Loss
Baseline	4.1927	4.8016	4.2332
No Scheduled Sampling	4.0106	3.5373	4.0234
No Label Smoothing	3.5029	4.0860	3.5738
Fine-tuning	3.6614	4.7512	4.2861

8 Discussion

The baseline provides the reference point for comparison. Removing scheduled sampling lowers all BLEU metrics, showing that scheduled sampling improves generalization during autoregressive caption generation.

Removing label smoothing also reduces BLEU performance, indicating that label smoothing contributes positively in this setup.

The best overall result is achieved by the fine-tuning configuration. The BLEU-4 score increases from 0.0472 in the baseline to 0.0649 with fine-tuning, making it the strongest final configuration in this study.

9 Qualitative Analysis

Predicted: man in the
Ground Truth: two ladies and three men looking at the ocean



Figure 1: Representative baseline output. The prediction is partially related to the scene but remains generic and incomplete relative to the reference caption.

Predicted: brown and white dog is running in the grass
Ground Truth: brown dog running over grass



Figure 2: Representative fine-tuned output. The generated caption correctly identifies the main object and action.

Predicted: hikes bread bread than war war so leading leading leading leading leading leading
leading leading leading leading leading leading leading leading
Ground Truth: four skiers walking up snow covered hill



Figure 3: Failure case. The output is repetitive and semantically unstable, illustrating an important residual limitation of the model.

10 Limitations

- All experiments were executed on CPU, resulting in long training times.
- Flickr8k is relatively small for image captioning.
- The work focuses on a classical CNN-LSTM architecture and does not compare with transformer-based alternatives.

11 Conclusion

This project implemented a complete CNN-LSTM image captioning pipeline on Flickr8k, covering preprocessing, vocabulary generation, supervised training, evaluation, and qualitative analysis. The final experiments show that scheduled sampling and label smoothing both improve performance relative to their corresponding ablations, while CNN fine-tuning provides the best final result. The repository now includes code, checkpoints, logs, figures, and results files suitable for a complete academic submission.

Available Submission Artifacts

- results/metrics_summary.csv
- results/experiment_summary.csv
- results/*_predictions.json
- logs/*

- figures/*
- LAB_LOG.md